

NORTHPORT PIER AT DEATHS DOOR, WI, USA

WMO: 994973

Lat: 45.2909N

Lon: 86.9768W

Elev: 188

StdP: 99.09

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.9	-16.5	N/A	N/A	N/A	N/A	N/A	N/A	15.2	-1.9	13.7	-5.4	4.1	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	27.6	N/A	25.7	N/A	24.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.4	11.7	10.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-6.0	-6.4	-0.8	4.6	10.4	16.2	20.6	20.1	16.6	10.2	3.5	-1.9
	DBStd	10.24	5.76	5.70	5.15	3.23	3.70	3.18	3.01	2.21	3.36	3.69	4.17	4.73
	HDD10.0	2102	495	460	335	162	41	1	0	0	1	43	197	368
	HDD18.3	4194	753	693	593	411	248	81	11	8	73	253	444	626
	CDD10.0	1132	0	0	1	1	52	186	329	313	197	50	2	0
	CDD18.3	182	0	0	0	0	1	15	83	63	19	1	0	0
	CDH23.3	531	0	0	0	0	2	42	323	127	36	1	0	0
	CDH26.7	89	0	0	0	0	0	5	66	14	5	0	0	0
(14) Wind	WSAvg	4.0	4.5	4.1	3.8	4.3	3.8	3.1	3.1	3.0	3.8	4.7	4.8	4.9
(15) Precipitation	PrecAvg	728	41	32	49	62	71	83	66	63	83	75	60	39
	PrecMax	942	113	63	127	110	118	221	179	102	198	180	121	75
	PrecMin	581	9	9	7	13	28	12	19	14	37	21	15	5
	PrecStd	87	28	13	33	29	27	51	39	25	37	39	28	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	5.7	14.9	15.2	22.9	27.1	31.2	28.5	26.7	21.6	13.7	9.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	5.1	3.4	10.7	12.6	20.0	24.3	28.7	26.4	24.2	18.5	11.9	6.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	3.4	2.1	7.8	10.9	18.2	22.5	26.7	24.7	22.5	16.8	10.4	5.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	1.8	0.8	5.7	9.2	16.3	21.0	25.0	23.5	21.1	15.4	9.3	4.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.3	6.6	5.8	7.0	6.6	6.5	5.7	5.9	4.8	4.5	4.6
		MCDBR	6.5	6.9	8.6	8.8	10.3	8.8	8.5	7.5	7.5	6.9	5.6	5.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.270	0.286	0.319	0.356	0.387	0.393	0.385	0.388	0.366	0.335	0.304	0.275
	taud		2.513	2.432	2.405	2.378	2.360	2.390	2.429	2.414	2.467	2.527	2.539	2.534
	Ebn at Noon		864	906	915	904	882	875	878	862	854	838	810	818
	Edh at Noon		65	85	101	113	120	117	111	109	95	78	64	58
(44) All-Sky Solar Radiation	RadAvg		1.33	2.31	3.57	4.62	5.64	6.07	6.24	5.28	4.05	2.46	1.41	1.03
	RadStd		0.13	0.26	0.33	0.43	0.45	0.47	0.26	0.35	0.30	0.26	0.14	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page